MATLAB | karthikkumarp2416sse-bit/5G-... | MIMO System BER Analysis | Ask Gemini

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Files

Name

- 5G AND 6G LAB
- LAB EXPERIMENT
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- ASSEMENT QNO 1.m
- EX : 2.m
- ex 1.m
- EX 2 ..m
- EX 2..m
- Ex 2.m
- ex 5.m
- ex1.m

Workspace

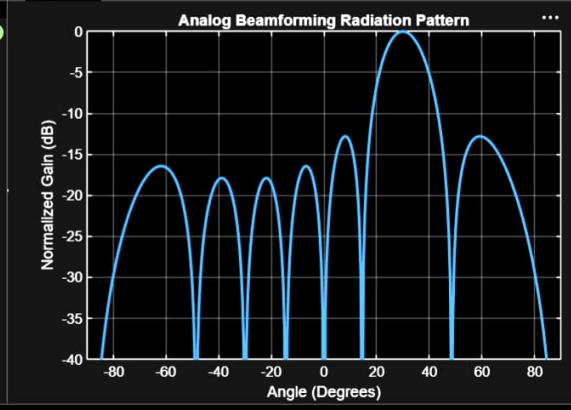
Name	Value	Size	Class
AF	1×361 double	1×361	double
k	361	1×1	double
N	8	1×1	double
phase	[0;1.5708;3.1...	8×1	double
theta	1×361 double	1×361	double
theta0	30	1×1	double

EX1.m | ex2.m | ex3.m | EX4.m | untitled4.m | EX6.m | EX7.m

/MATLAB Drive/EX7.m

```
9   % Steering Vector
10  AF = zeros(size(theta));
11
12  for k = 1:length(theta)
13
14      phase = pi*(0:N-1)'*(sind(theta(k)) - sind(theta0));
15      AF(k) = abs(sum(exp(1j*phase)));
16
17  end
18
19  % Normalize Array Factor
20  AF = AF/max(AF);
21
22  % Plot Beam Pattern
23  figure;
24  plot(theta,20*log10(AF),'LineWidth',2);
25  grid on;
26  axis([-90 90 -40 0]);
27
28  xlabel('Angle (Degrees)');
29
```

Figure 1



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